Pengfei Hu

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Education

Stevens Institute of Technology Sep. 2023 – Present
Ph.D. in Computer Science *Hoboken, NJ*

- Advisor: Dr. Yue Ning; GPA: 3.95 / 4.00;
- **Research Interest:** Transfer Learning, Post-Training, Large Language Model, Graph Learning, Interpretability.

University of Southern California Sep. 2021 – May 2023
M.S. in Analytics *Los Angeles, CA*

- GPA: 4.00 / 4.00;
- Research Assistant advised by Dr. Sze-chuan Suen working on monitoring of diabetes via wearable devices.

Nanjing University of Finance and Economics Sep. 2016 – Jun. 2020
B.S. in Economic Statistics *Nanjing, China*

- GPA: 3.63 / 4.00 (Rank 1st/150 in Major, National Scholarship).
- Working on relative deprivation in the digital era (one publication) supervised by Dr. Qinghai Li.
- Interned at *Alibaba Group* (2021), *Tencent* (2020), *Didi Global* (2019–2020), and *PwC Consulting* (2019).

Internship Experience

Amazon Web Services Sep. 2026 – Dec. 2026
Applied Scientist Intern *New York City, NY*

- **Mentor:** *Sébastien Jean*; **Manager:** *James Gung*;
- **Topic:** Evaluating and Understanding memory system for proactive agents in long-horizon personal workflow.

Oak Ridge National Laboratory May 2025 – Aug. 2025
Research Scientist Intern *Oak Ridge, TN*

- **Mentor:** *Ming Fan*; **Leader:** *Dan Lu*;
- **Project 1: Adaptive Graph Transformer for Multi-Reservoir Inflow Forecasting**
  - Designed an adaptive graph learning module that captures latent inter-reservoir dependencies, coupled with a Transformer encoder to model spatiotemporal inflow dynamics in the Upper Colorado River Basin;
  - **Paper: First-author** workshop paper at IEEE International Conference on Data Mining (ICDM) 2025; extended version (**co-first-author**) published in Environmental Modelling & Software (IF: 4.6).
- **Project 2: Hydrological Domain-Conditioned Modulation for Cross-Reservoir Generalization**
  - Conditioned a shared backbone on reservoir meta-attributes (geography, storage, climate) via feature-wise linear modulation for zero-shot transfer to unseen reservoirs;
  - **Paper: First-author** Oral at AAAI 2026 Workshop on AI for Environmental Science.

Selected Projects

Tool-Integrated Reinforcement Learning for Emergency Care Prediction
Under review

- Built a Qwen3-4B agent for emergency-care prediction, implementing multi-turn tool-use training with GRPO in verl and verl-tool, with vLLM for rollout inference.
- Mapped gradient-boosted tree feature importance to clinical functions, designing function-matching and risk rewards to improve evidence selection and numerical risk estimation.
- Distilled clinical skills through a reverse curriculum that gradually removes prompt guidance during RL.

Transparent Domain Adaptation via Concept-Grounded Orthogonal Inference
ICML 2026

- Proposed an orthogonal and concept-grounded inference method for transparent adaptation in clinical data.
- Designed a sparse autoencoder with a dictionary-induced metric to factorize patient representations into label-invariant and domain-covariant components via M-orthogonal residual projection.
- Provided human-understandable explanations via sparse-dimension ablation while achieving superior performance over state-of-the-art DA baselines on two EHR datasets under spatial, temporal, and facility distribution shifts.

LLM Agent for Causal Discovery in Diagnosis Prediction

EMNLP 2025 Findings

- Designed an agent-based causal discovery framework leveraging large language models to explore disease relationships and improve interpretability and predictive accuracy in clinical diagnosis tasks.
- Constructed a domain knowledge vector database using ChromaDB, integrating biomedical knowledge from Wikipedia and enhancing reasoning coverage via RAG over structured knowledge.
- Implemented the agentic reasoning pipeline using LangChain and evaluated performance on public EHR datasets.

Skills

Foundation Models & Reasoning Techniques: LLM Post-Training (SFT, PPO, GRPO), Test-Time Reasoning, On-Policy Distillation, LLM Agents, Tool-Integrated Reasoning, Retrieval-Augmented Generation (RAG).

Infrastructure & Scalable ML Systems: PyTorch, Hugging Face Transformers, verl, verl-tool, TRL, Accelerate, SGLang, vLLM, PyTorch Geometric, TensorFlow, TensorBoard, Weights & Biases, AWS (EC2, S3).

Agent & Data Infrastructure: LangChain, ChromaDB.

Model Architectures: Transformers, Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Sparse Auto Encoder.

Programming Languages: Python, R, SQL, Java, Latex.

Selected Publications

(The Google Scholar link can be found here, * equal contribution)

First-authored Publications:

1. **Pengfei Hu**, et al., "Incentivizing Tool-Use Reinforcement Learning for Post-Arrival Emergency Care Prediction", *Under review*, 2026.
2. **Pengfei Hu**, et al., "Exploring Accurate and Transparent Domain Adaptation in Predictive Healthcare via Concept-Grounded Orthogonal Inference", *International Conference on Machine Learning* (**ICML**), 2026.
3. **Pengfei Hu**, et al., "Discovering Hierarchy-Grounded Domains with Adaptive Granularity for Clinical Domain Generalization", *International Conference on Information and Knowledge Management* (**CIKM**), 2026. **Oral.**
4. **Pengfei Hu**, et al., "Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning", *IEEE Journal of Biomedical and Health Informatics* (**IEEE JBHI**, IF: 6.8), 2026.
5. **Pengfei Hu***, et al., "AGFormer: Adaptive Spatiotemporal Graph Informed Transformer for Multi-Reservoir Inflow Forecasting", *Environmental Modelling & Software* (IF: 4.6), 2026.
6. **Pengfei Hu***, et al., "MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records", *IEEE International Conference on Big Data* (**IEEE BigData 2024**).

Collaborative Publications:

1. Xiaoxue Han, **Pengfei Hu**, et al., "Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery", *Findings of the Association for Computational Linguistics* (**EMNLP 2025 Findings**).
2. Hsin-Ling Hsu, ..., **Pengfei Hu**, et al., "MEDPLAN: A Two-Stage RAG-Based System for Personalized Medical Plan Generation", *Annual Meeting of the Association for Computational Linguistics* (**ACL 2025**, Industry Track).

Honors

- Best Doctoral Research Award at the iCNS @ Stevens AI Engineering and Science Symposium *Spring 2026*
- Stevens Institute of Technology Provost Doctoral Fellowship *2023 – 2028*
- University Outstanding Graduate (Summa Cum Laude) *Spring 2020*
- National Scholarship of Zhongli International (Top 0.5% in school) *Fall 2019*